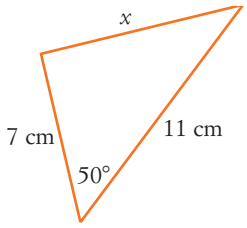
WORKSHEET

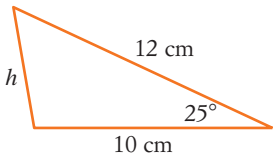
Chapter 10: The cosine rule – Angles and sides

Using the cosine rule, calculate the missing lengths and angles in these right-angled triangles. Give answers to one decimal place or to the nearest degree,

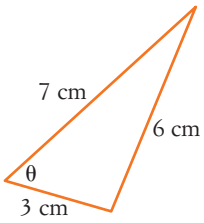
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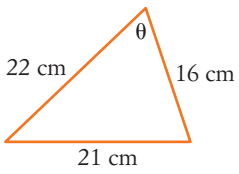
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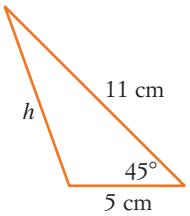
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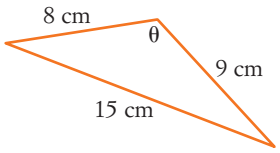
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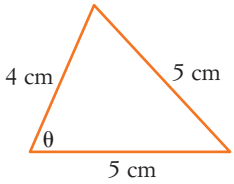
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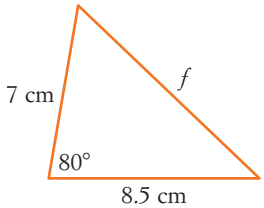
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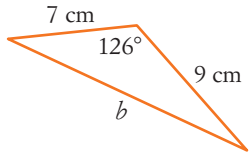


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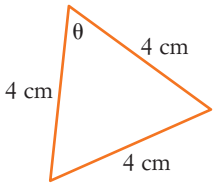


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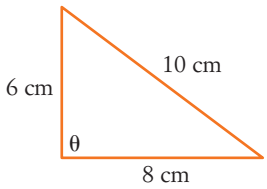
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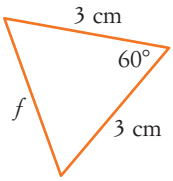
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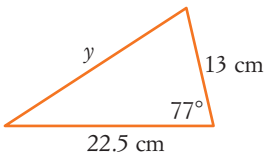
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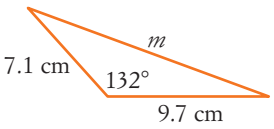
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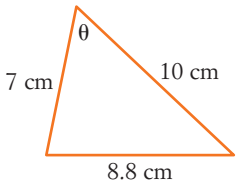
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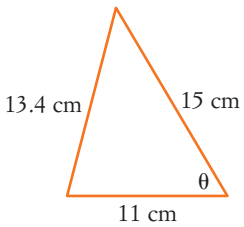
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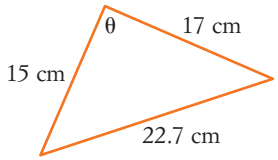
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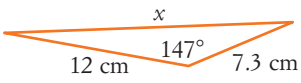
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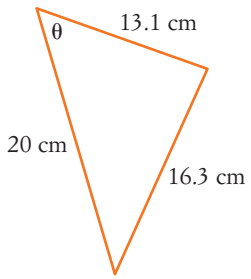


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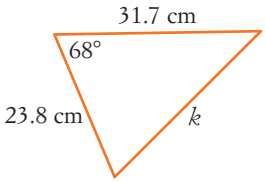


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19



20



21 In $\triangle ABC$, $\angle A = 75°$, $b = 14$ cm and $c = 18$ cm.

i Draw a sketch of $\triangle ABC$, showing all the information given.

ii Calculate the length of a to one decimal place.

22 In $\triangle XYZ$, $x = 9.4$ cm, $y = 6.2$ cm and $z = 8.1$ cm.

i Draw a sketch of $\triangle ABC$, showing all the information given.

ii Calculate the size of $\angle X$ to the nearest degree.

23 In $\triangle DEF$, $\angle E = 136°$, $d = 5$ cm and $f = 7.2$ cm.

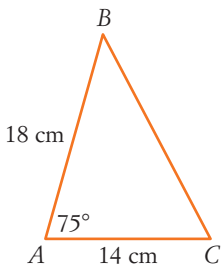
i Draw a sketch of $\triangle DEF$, showing all the information given.

ii Calculate the length of e to one decimal place.

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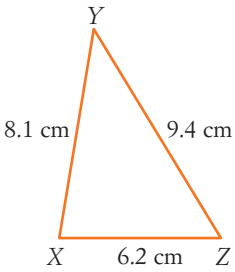
Answers

1 8.4 cm
2 5.1 cm
3 58°
4 65°
5 8.3 cm
6 124°
7 66°
8 8.5 cm
9 14.3 cm
10 60°
11 90°
12 3 cm
13 23.3 cm
14 15.4 cm
15 59°
16 60°
17 90°
18 18.6 cm
19 54°
20 23.8 cm
21 **i**



ii 14 cm

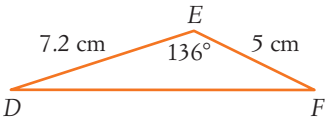
22 **i**



ii 81°

23

i



ii 11.3 cm

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